Music Genre Classifier

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Introduction

In the last few decades, streaming platforms have revolutionized access to music, with more music being easily accessible now than at any other time in history. Complicated ML algorithms underlie all of these platforms, opening up new pathways for artists to achieve increased discovery and revenue. One such algorithm that Spotify, for example, relies on is its genre classification and tagging system. While Spotify employees are mostly responsible for monitoring the creation of new genres as listening habits and musical categories evolve over time, Spotify has an algorithm that classifies songs into these genres. We set out to replicate this genre classification system, to gain a deeper understanding of how algorithms can be used to shape musical taste and discovery.

Methodology

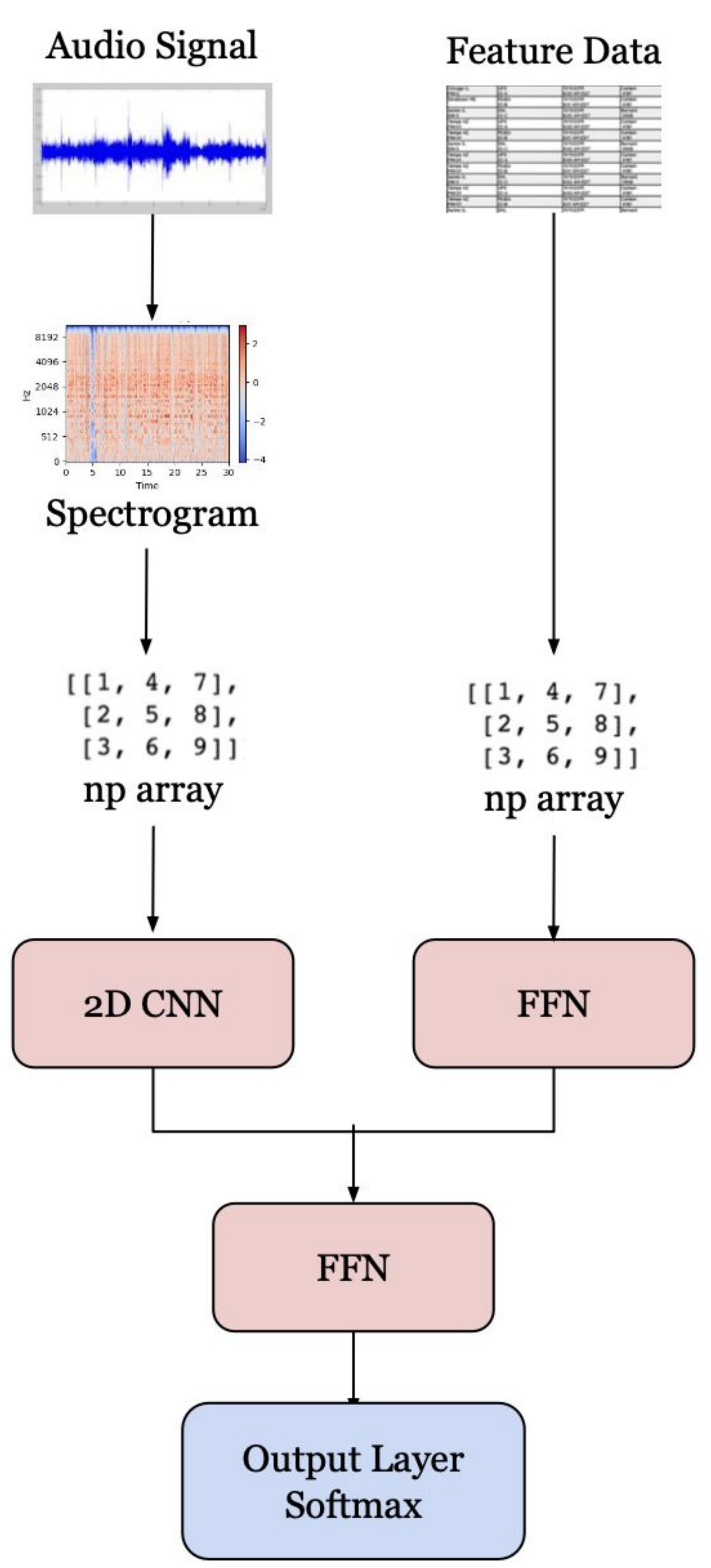


Figure 1. Algorithm flowchart.

The model paper suggested using a pre-trained CNN to analyze spectrographic data and separate machine learning algorithms (SVMs, XGB, random forest) to analyze feature data. We combined these techniques into one pipeline by continuing to use a 2D CNN for the spectrographic data, using an MLP with Dropout for our feature data, and concatenating the two before using a final dense output layer with softmax activation. For comparison, we also used solely the spectrographic data with a CNN to see if this would yield better results (as in the paper). We used trial-and-error to tune our hyperparameters, such as the learning rate and embedding size, and used general observation to select the number of epochs for training.

Data

The model paper used the dataset *Audio Set,* which contained 40,000 10–second audio files divided into 7 genres. This dataset was not easily accessible, so we have chosen to use *Free Music Archive (FMA)*, a dataset with 8,000 elements—1000 30-second audio files for eight genres, including a test (10%) and validation (10%) set. For each audio file, the dataset contained a spectrogram, feature data, and its genre classification. Feature data included qualities such as danceability, acousticness, speechiness, and tempo.

For consistency, in our preprocessing stage, we converted the audio files into a spectrogram using the librosa library. Comparisons between the dataset's and our generated spectrogram are shown below.

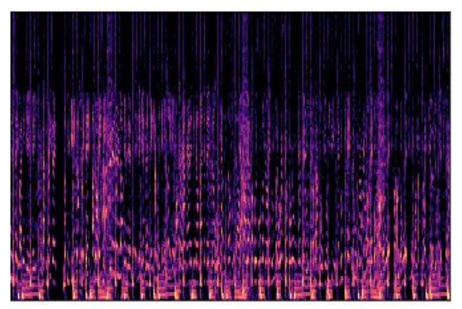
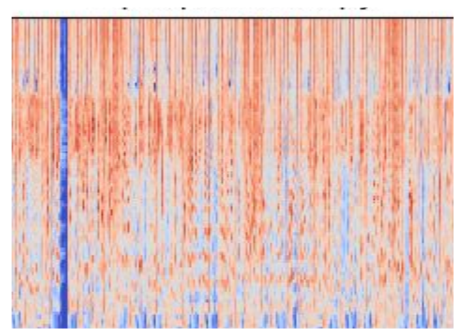
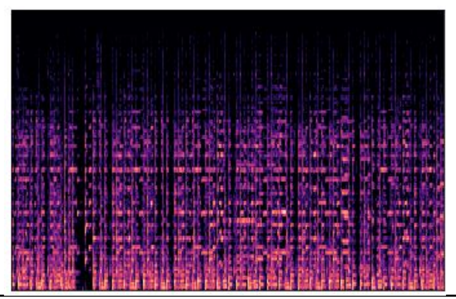
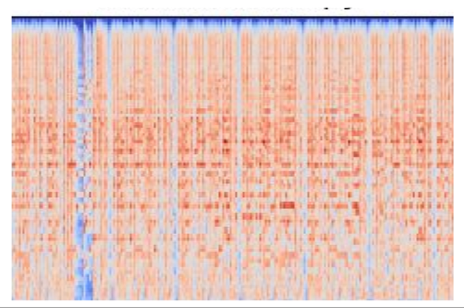
Genre	FMA Spectrogram	Generated Spectrogram
Hip Hop		
Blues		

Figure 2. FMA vs. Generated Spectrograms

Results

Our best accuracy of 51% was achieved by a CNN model combined with an MLP, with optimally tuned hyperparameters. All results are given below. We assigned class labels (0-7) as one-hot vectors for the 8 classes. and calculated the AUC from the ROC.

Model	Accuracy	AUC
2D CNN	0.49	0.68
2D CNN + Tabular Data MLP	0.51	0.70
Paper's Fine-Tuned 2D CNN	0.64	0.89

Figure 3. Model result metrics.

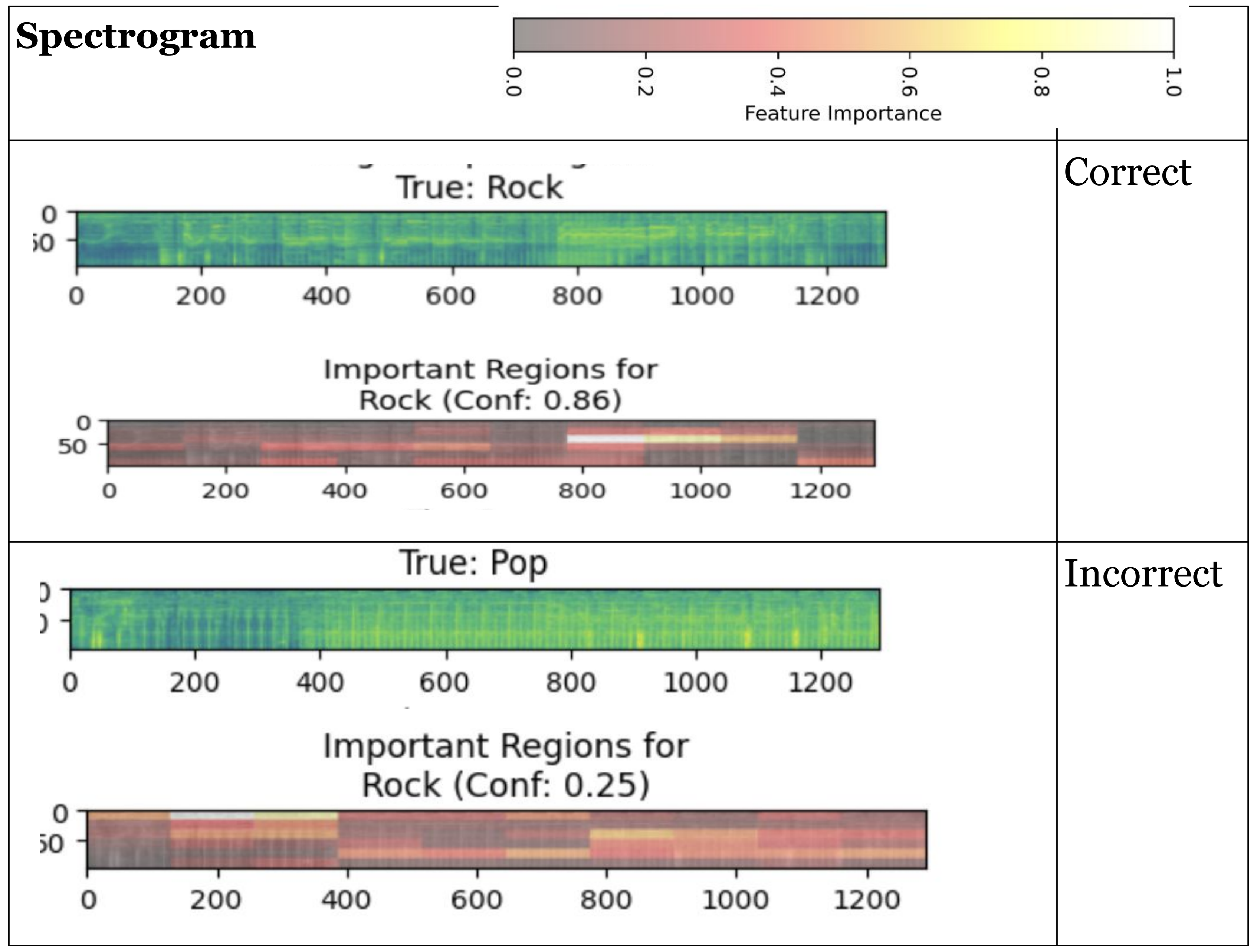


Figure 4. LIME Explanations

Discussion

It is difficult to classify music into genres. Simply put, eight genres is far too few to sort music into with any certainty. The average song will often combine many elements from across diverse genres, which probably added noise to the model and limited its success. In addition, the definition of a genre shifts often. What was considered "pop" forty years ago might be called "new wave" today, or in our experiment might have been sorted into "rock". In a classification landscape that relies on subjectivity to define categories, it seems difficult for ML models to keep up, especially with our lack of access to sophisticated hardware and data. This is why, for example, Spotify employs many musicologists, who can help keep sorting algorithms aligned with trends in the music industry. They also have a much richer song database that helps their models achieve higher performance.

Future work could involve expanding data access. Working directly with Spotify's API could be a fruitful approach, scraping more tabular song data. For example, every artist on Spotify's API has a list of associated genres; this could be used as additional input, or perhaps as a way to find similarity between genres. Having access to more compute (outside of Google Colab)would also help our model be able to achieve deeper levels of training, perhaps finding more complex patterns in the data that helps it achieve better classification scores.